BANK LOAN ANALYSIS QUERIES

PROBLEM STATEMENT

1.TOTAL LOAN APPLICATIONS:

SELECT COUNT(id) AS Total_Loans_Applications

FROM bank_loan_analysis;

OUTPUT:

Result Grid | Filter Rows:

Total_Loans_Applications
38576

2.TOTAL AMOUNT FUNDED:

SELECT sum(loan_amount) AS Total_Funded_Amount

FROM bank_loan_analysis;

OUTPUT:

Result Grid | Filter Rows:

Total_Funded_Amount
435757075

3. TOTAL AMOUNT RECEIVED:

SELECT SUM(total_payment) AS Total_Amount_Received

FROM bank_loan_analysis;

OUTPUT:

Result Grid | Filter Rows:

Total_Amount_Received
473070933

4. AVERAGE INTEREST RATE:

SELECT ROUND(AVG(int_rate),2) AS Average_Interest_Rate

FROM bank_loan_analysis;

OUTPUT:

	Average_Interest_Rate
▶	0.12

5. AVERAGE DEBT TO INCOME RATIO (DTI):

SELECT ROUND(AVG(dti),2) AS Average_DTI

FROM bank_loan_analysis

OUTPUT:

	Average_DTI
▶	0.13

6.GOOD LOAN ISSUED.

A. GOOD LOAN PERCENTAGE:

SELECT

CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'current' THEN id

END)*100/COUNT(id),2),'%') AS Good_Loan_Percentage

FROM bank_loan_analysis;

OUTPUT:

	Good_Loan_Percentage
▶	86.18%

B. GOOD LOAN APPLICATIONS:

SELECT COUNT(id) AS Good_Loan_Application

FROM bank_loan_analysis

WHERE loan_status = 'Fully Paid' OR loan_Status = 'current'

OUTPUT:

	Good_Loan_Application
▶	33243

C. GOOD LOAN FUNDED AMOUNT:

SELECT SUM(loan_amount) AS Good_Funded_Amount

FROM bank_loan_analysis

WHERE loan_status = 'Fully Paid' OR loan_Status = 'current'

OUTPUT:

Result Grid | Filter Rows:

Good_Funded_Amount
370224850

D. GOOD LOAN RECEIVED AMOUNT:

SELECT SUM(total_payment) AS Good_Loan_Amount_Received

FROM bank_loan_analysis

WHERE loan_status = 'Fully Paid' OR loan_Status = 'current'

OUTPUT:

Result Grid | Filter Rows:

Good_Loan_Amount_Received
435786170

7.BAD LOAN ISSUED.

A. BAD LOAN PERCENTAGE:

SELECT

CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END)*100/COUNT(id),2),'%')

AS Bad_Loan_Percentage

FROM bank_loan_analysis;

OUTPUT:

Result Grid | Filter Rows:

Bad_Loan_Percentage
13.82%

B. BAD LOAN APPLICATIONS:

SELECT COUNT(id) AS Bad_Loan_Application

FROM bank_loan_analysis

WHERE loan_status = 'Charged Off'

OUTPUT:

Result Grid | Filter Rows:

Bad_Loan_Application
5333

C. BAD LOAN FUNDED AMOUNT:

SELECT SUM(loan_amount) AS Bad_Funded_Amount

FROM bank_loan_analysis

WHERE loan_status = 'Charged Off'

OUTPUT:

Result Grid | Filter Rows:

Bad_Funded_Amount
65532225

D. BAD LOAN RECEIVED AMOUNT:

SELECT SUM(total_payment) AS Bad_Loan_Amount_Received

FROM bank_loan_analysis

WHERE loan_status = 'Charged Off'

OUTPUT:

Result Grid | Filter Rows:

Bad_Loan_Amount_Received
37284763

8. LOAN STATUS:

SELECT loan_status,

COUNT(id) AS Total_ID,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Recieved,

ROUND(AVG(int_rate),2) AS Average_Intrest_Rate,

ROUND(AVG(dti),2) AS Average_DTI

FROM bank_loan_analysis

GROUP BY loan_status;

OUTPUT:

loan_status	Total_ID	Total_Funded_Amount	Total_Amount_Recieved	Average_Intrest_Rate	Average_DTI
Charged Off	5333	65532225	37284763	0.14	0.14
Fully Paid	32145	351358350	411586256	0.12	0.13
Current	1098	18866500	24199914	0.15	0.15

9.MONTHLY LOAN APPLICATION TREND:

```
SELECT
    YEAR(issue_date) AS Loan_Year,
    MONTH(issue_date) AS Loan_Month_Number,
    MONTHNAME(issue_date) AS Loan_Month,
    COUNT(*) AS Total_Applications
FROM bank_loan_analysis
GROUP BY
    YEAR(issue_date),
    MONTH(issue_date),
    MONTHNAME(issue_date)
ORDER BY Loan_Year,Loan_Month_Number;
```

OUTPUT:

Loan_Year	Loan_Month_Number	Loan_Month	Total_Applications
2021	1	January	2332
2021	2	February	2279
2021	3	March	2627
2021	4	April	2755
2021	5	May	2911
2021	6	June	3184
2021	7	July	3366
2021	8	August	3441
2021	9	September	3536
2021	10	October	3796
2021	11	November	4035
2021	12	December	4314

10.STATEWISE LOAN APPLICATION TREND:

```
SELECT
address_state,
COUNT(id) AS Total_Loan_Application,
SUM(loan_amount) AS Total_Funded_Amount,
```

```
SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_analysis
GROUP BY address_state
ORDER BY address_state;
```

OUTPUT:

address_state	Total_Loan_Application	Total_Funded_Amount	Total_Amount_Received
AK	78	1031800	1108570
AL	432	4949225	5492272
AR	236	2529700	2777875
AZ	833	9206000	10041986
CA	6894	78484125	83901234
CO	770	8976000	9845810
CT	730	8435575	9357612
DC	214	2652350	2921854
DE	110	1138100	1269136
FL	2773	30046125	31601905
GA	1355	15480325	16728040
HI	170	1850525	2080184
IA	5	56450	64482
ID	6	59750	65329
IL	1486	17124225	18875941
IN	9	86225	85521
KS	260	2872325	3247394
KY	320	3504100	3792530
LA	426	4498900	5001160
MA	1310	15051000	16676279
MD	1027	11911400	12985170
ME	3	9200	10808
MI	685	7829900	8543660
MN	592	6302600	6750746
MO	660	7151175	7692732
MS	19	139125	149342
MT	79	829525	892047
NC	759	8787575	9534813
OR	436	4720150	4966903
PA	1482	15826525	17462908
RI	196	1883025	2001774
SC	464	5080475	5462458
SD	63	606150	656514
TN	17	162175	141522
TX	2664	31236650	34392715
UT	252	2849225	2952412
VA	1375	15982650	17711443
VT	54	504100	534973
WA	805	8855525	9531739
WI	446	5070450	5485161
WV	167	1830525	1991936
WY	79	890750	1046050

11.LOAN PURPOSE ANALYSIS:

```
SELECT purpose,
count(*) AS Total_Applications,
SUM(loan_Amount) AS Total_Funded_Amount FROM bank_loan_analysis
GROUP BY purpose
ORDER BY Total_Applications DESC;
```

OUTPUT:

Result Grid | Filter Rows: | Export:

purpose	Total_Applications	Total_Funded_Amount
Debt consolidation	18214	232459675
credit card	4998	58885175
other	3824	31155750
home improvement	2876	33350775
major purchase	2110	17251600
small business	1776	24123100
car	1497	10223575
wedding	928	9225800
medical	667	5533225
moving	559	3748125
house	366	4824925
vacation	352	1967950
educational	315	2161650
renewable_energy	94	845750

12. HOME OWNERSHIP ANALYSIS:

```
SELECT
home_ownership AS Home_Ownership,
COUNT(*) AS Total_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_analysis
GROUP BY home_ownership
ORDER BY Total_Applications DESC;
```

OUTPUT:

Home_Ownership	Total_Applications	Total_Funded_Amount	Total_Amount_Received
RENT	18439	185768475	201823056
MORTGAGE	17198	219329150	238474438
OWN	2838	29597675	31729129
OTHER	98	1044975	1025257
NONE	3	16800	19053

13. GRADE – WISE LOAN ANALYSIS:

```
SELECT grade,
COUNT(*) AS Total_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_analysis
GROUP BY grade
ORDER BY grade;
```

OUTPUT:

grade	Total_Applications	Total_Funded_Amount	Total_Received_Amount
A	9689	84252225	88051563
B	11674	130703975	140775015
C	7904	87456450	95973518
D	5182	63920800	70823891
E	2786	44165100	49164151
F	1028	18910450	21016738
G	313	6348075	7266057

14. RANK STATE BY TOTAL FUNDED AMOUNT:

```
SELECT address_state,
COUNT(*) AS Total_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
RANK () OVER (ORDER BY SUM(loan_amount) DESC) AS State_Rank
FROM bank_loan_analysis
GROUP BY address_state;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell C

address_state	Total_Applications	Total_Funded_Amount	State_Rank
CA	6894	78484125	1
NY	3701	42077050	2
TX	2664	31236650	3
FL	2773	30046125	4
NJ	1822	21657475	5
IL	1486	17124225	6
VA	1375	15982650	7
PA	1482	15826525	8
GA	1355	15480325	9
MA	1310	15051000	10
OH	1188	12991375	11
MD	1027	11911400	12
AZ	833	9206000	13
CO	770	8976000	14
WA	805	8855525	15
NC	759	8787575	16
CT	730	8435575	17
MI	685	7829900	18
MO	660	7151175	19
MN	592	6302600	20
NV	482	5307375	21
SC	464	5080475	22
WI	446	5070450	23
AL	432	4949225	24
OR	436	4720150	25
LA	426	4498900	26
KY	320	3504100	27
OK	293	3365725	28
OK	293	3365725	28
KS	260	2872325	29
UT	252	2849225	30
DC	214	2652350	31
AR	236	2529700	32
NH	161	1917900	33
NM	183	1916775	34
RI	196	1883025	35
HI	170	1850525	36
WV	167	1830525	37
DE	110	1138100	38
AK	78	1031800	39
WY	79	890750	40
MT	79	829525	41
SD	63	606150	42
VT	54	504100	43
TN	17	162175	44
MS	19	139125	45
IN	9	86225	46
ID	6	59750	47
IA	5	56450	48
NE	5	31700	49
ME	3	9200	50

15.CATEGORISE LOAN USING CASE:

SELECT

CASE WHEN loan_amount< 5000 THEN 'Small Loan'

WHEN loan_amount>5000 AND loan_amount<100000 THEN 'Medium Loan'

ELSE 'Large Loan'

END AS Loan_Category,

COUNT(*) AS Total_Loans,

SUM(loan_Amount) AS Total_Funded_Amount

FROM bank_loan_analysis

GROUP BY Loan_Category

ORDER BY Total_Funded_Amount DESC;

OUTPUT:

Result Grid | Filter Rows: | Export:

Loan_Category	Total_Loans	Total_Funded_Amount
Medium Loan	29463	403626875
Small Loan	7136	22245200
Large Loan	1977	9885000

16.EACH LOAN PURPOSE PERCENTAGE CONTRIBUTION:

SELECT

purpose,

SUM(loan_amount) AS Total_Funded_Amount,

ROUND(SUM(loan_amount)*100/SUM(SUM(loan_amount)) OVER(),2) AS Percentage_Contribution

FROM bank_loan_analysis

GROUP BY purpose

ORDER BY Percentage_Contribution DESC;

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap

purpose	Total_Funded_Amount	Percentage_Contribution
Debt consolidation	232459675	53.35
credit card	58885175	13.51
home improvement	33350775	7.65
other	31155750	7.15
small business	24123100	5.54
major purchase	17251600	3.96
car	10223575	2.35
wedding	9225800	2.12
medical	5533225	1.27
house	4824925	1.11
moving	3748125	0.86
educational	2161650	0.50
vacation	1967950	0.45
renewable_energy	845750	0.19